

REACTJS (ASSIGNMENT 9)

Aim

To create a React application named **cricketapp** that demonstrates the use of **ES6 features** such as **map()**, **arrow functions**, **destructuring**, and the **spread operator** by displaying a list of cricket players, filtering players based on their scores, separating odd and even team players, and merging multiple player arrays using React components.

1. Create a new React application named cricketapp.

`npx create-react-app cricketapp`

```
PS C:\Users\eshit\OneDrive\Desktop> npx create-react-app cricketapp

To address issues that do not require attention, run:
  npm audit fix

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.

Created git commit.

Success! Created cricketapp at C:\Users\eshit\OneDrive\Desktop\cricketapp
Inside that directory, you can run several commands:

  npm start
    Starts the development server.

  npm run build
    Bundles the app into static files for production.

  npm test
    Starts the test runner.

  npm run eject
    Removes this tool and copies build dependencies, configuration files
    and scripts into the app directory. If you do this, you can't go back!

We suggest that you begin by typing:

  cd cricketapp
  npm start

Happy hacking!
```

2. Navigate to the project folder.

`cd cricketapp`

3. App.js

`import './App.css';`

`import ListofPlayers from './Components/ListofPlayers';`

`import IndianPlayers from './Components/IndianPlayers';`

```

function App() {

  const flag = true;

  if (flag) {
    return (
      <div className="App">
        <ListofPlayers />
      </div>
    );
  }
  else {
    return (
      <div className="App">
        <IndianPlayers />
      </div>
    );
  }
}

```

```

export default App;

```

4. Components/ ListofComponents.js

```

import React from "react";

```

```

function ListofPlayers() {

```

```

  const players = [

```

```
{ name: "Virat Kohli", score: 95 },
{ name: "Rohit Sharma", score: 82 },
{ name: "Shubman Gill", score: 68 },
{ name: "KL Rahul", score: 75 },
{ name: "Hardik Pandya", score: 63 },
{ name: "Ravindra Jadeja", score: 72 },
{ name: "R Ashwin", score: 54 },
{ name: "Mohammed Shami", score: 49 },
{ name: "Jasprit Bumrah", score: 85 },
{ name: "Mohammed Siraj", score: 66 },
{ name: "Kuldeep Yadav", score: 70 }
];

const below70 = players.filter(player => player.score < 70);

return (
  <div>

    <h1>List of Players</h1>

    {players.map((player, index) => (
      <div key={index}>
        <h3>
          {player.name} - {player.score}
        </h3>
      </div>
    ))}
  )}
```

```
<hr />
```

```
<h2>Players with Score Below 70</h2>
```

```
{below70.map((player, index) => (
```

```
  <div key={index}>
```

```
    <h3>
```

```
      {player.name} - {player.score}
```

```
    </h3>
```

```
  </div>
```

```
  )})
```

```
</div>
```

```
);
```

```
}
```

```
export default ListofPlayers;
```

5. Components/ IndianPlayers.js

```
import React from "react";
```

```
function IndianPlayers() {
```

```
  const players = [
```

```
    "Virat",
```

```
    "Rohit",
```

```
    "Gill",
```

```
    "Rahul",
```

```
    "Hardik",  
    "Jadeja"  
];
```

```
const [  
    odd1,  
    even1,  
    odd2,  
    even2,  
    odd3,  
    even3  
] = players;
```

```
const T20Players = [  
    "Surya",  
    "Tilak",  
    "Rinku"  
];
```

```
const RanjiPlayers = [  
    "Pujara",  
    "Rahane",  
    "Sarfraz"  
];
```

```
const mergedPlayers = [  
    ...T20Players,  
    ...RanjiPlayers
```

```
];
```

```
return (
```

```
  <div>
```

```
    <h1>Odd Team Players</h1>
```

```
    <h3>{odd1}</h3>
```

```
    <h3>{odd2}</h3>
```

```
    <h3>{odd3}</h3>
```

```
    <hr />
```

```
    <h1>Even Team Players</h1>
```

```
    <h3>{even1}</h3>
```

```
    <h3>{even2}</h3>
```

```
    <h3>{even3}</h3>
```

```
    <hr />
```

```
    <h1>Merged Players</h1>
```

```
    {mergedPlayers.map((player, index) => (
```

```
      <h3 key={index}>{player}</h3>
```

```
    )}}
```

</div>

);

}

export default IndianPlayers;

6. Output

List of Players

Virat Kohli - 95
Rohit Sharma - 82
Shubman Gill - 68
KL Rahul - 75
Hardik Pandya - 63
Ravindra Jadeja - 72
R Ashwin - 54
Mohammed Shami - 49
Jasprit Bumrah - 85
Mohammed Siraj - 66
Kuldeep Yadav - 70

Players with Score Below 70

Shubman Gill - 68
Hardik Pandya - 63
R Ashwin - 54
Mohammed Shami - 49
Mohammed Sirai - 66

7. Result

The React application was developed successfully. The list of cricket players was displayed using the `map()` method, players scoring below 70 were filtered using arrow functions, odd and even team players were displayed using destructuring, and the T20 and Ranji Trophy player arrays were merged using the spread operator.

8. Conclusion

The experiment was completed successfully. A React application named `cricketapp` was created to demonstrate the use of ES6 features, including `map()`, arrow functions, destructuring, and the spread operator. Components were rendered conditionally using a simple if-else statement based on a flag variable.